

ELEC451: POWER ELECTRONICS**LAB 0: INTRODUCTION TO SIMULATIONS AND EXPERIMENTAL INSTRUMENTS****OBJECTIVES**

1. Learn how to properly use an electronic circuit simulation software (PSIM), its key control parameters (time step, total time, print time), and the main curves and measurements (time domain curves, FFT, THD, cursors, etc.)
2. Learn how to properly use the oscilloscope (configure the scale, the trigger, measurements, captures, cursors, FFT) and configure and calibrate the different probes (standard probe, isolated probe, current probe)
3. Understand the structure and requirements of a lab report and the requirements for oscilloscope/simulation captures

LAB INSTRUCTIONS

The lab is divided into two sections: simulation and experiments. During the first section, the simulations are performed, whereas in the second section the experimental activities are conducted.

For **LAB 0 ONLY**, after completing the first lab section each group must answer the Lab 0 Quiz posted on Canvas. For this reason, it is highly recommended to read carefully and complete the entire lab instructions, since you will not have access to the lab to repeat the experiments.

Report due date: one week after lab session.

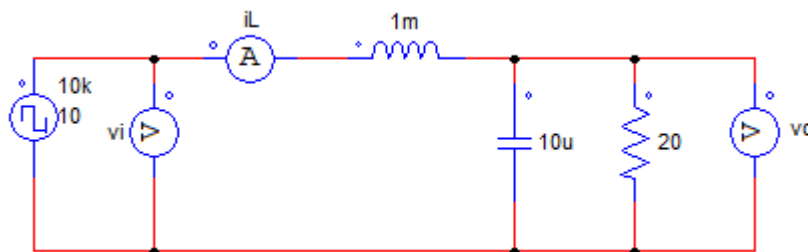
Required Components:

- PSIM simulation software
- Oscilloscope, regular probes, isolated probes, current probes

READ THE ENTIRE GUIDE BEFORE PROCEEDING!**SIMULATIONS****ACTIVITY 1 – CIRCUIT SETUP**

- a. Build the following circuit using PSIM and set the parameters for each element as detailed on the table.

When setting the parameters inside a block it is recommended to use scientific notation, e.g. $C = 10\text{e-}6$ or using prefixes $C = 10\text{u}$ (avoid entering $C = 0.000010$). Also, check the boxes to display the parameters on the circuit schematic, such as the names of the probes and the values of the components.



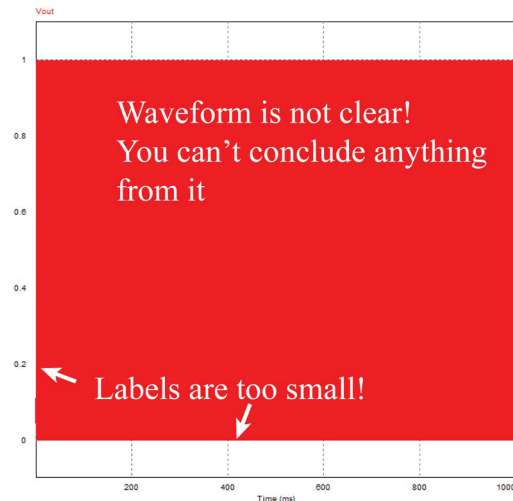
Parameter	Value
Square-wave Source Voltage/Frequency	10 V / 10 kHz
Inductance	1 mH
Capacitance	10 μF
Resistance	20 Ω

- b. Place the voltage and current probes paying attention to the position of the positive node (small circle). Give the probes proper names.
- c. Run the simulation (*Simulate/Run Simulation*).

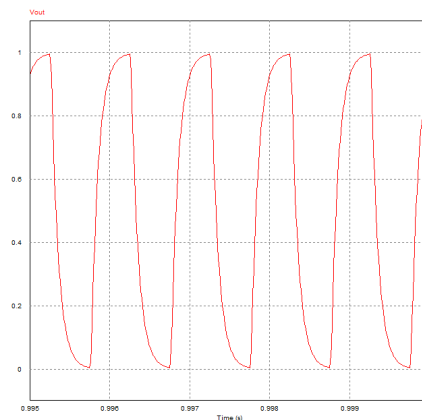
ACTIVITY 2 – CAPTURES

- a. Plot the three waveforms obtained from the simulations in separate screens (*Screen/Add Screen*).
- b. Adjust the aspect ratio of the windows to make them suitable for the format of the lab report. Remember that what you want to show must be visible!
- c. Be sure that the size of the labels of the final figure is large enough. You can configure the font on the *Screen* tab under the *Properties* menu (*Screen/Add Screen*).
- d. The thickness of the curves can also be configured on the *Curves* tab under the *Properties* menu.
- e. Copy the figure (*Edit/Copy to Clipboard*). Paste the figure on your report and make sure the aspect ratio, labels size or curve thickness are right. Repeat b), c) or d) if necessary. Remember that **all** of your figure must be legible. Below is an example of what is wrong and what is expected:

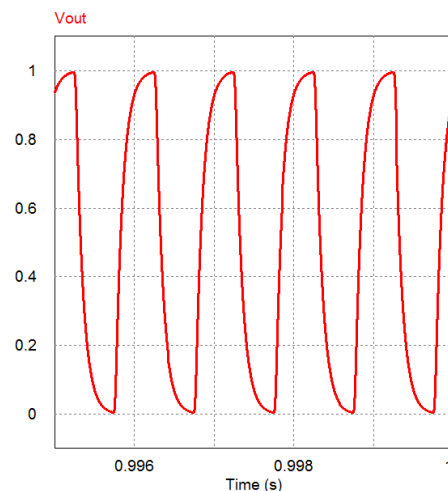
- i. **Wrong:** In this figure, the labels are too small, and the plot does not display the properties of the waveform you may want to show.



- ii. **Wrong:** In this figure, the labels are too small, and even though the plot displays the properties of the waveform you want to show, the line thickness is not adequate.



- iii. **Right:** In this figure, the labels are properly sized, and the waveform is clearly visible.



ACTIVIY 3 – STEADY-STATE VS. TRANSIENT

- Add the *Simulation Control* tool to your schematic (*Simulate/Simulation Control*). Set the *Time Step* to 1 μ s and the *Total Time* to a proper value in order to show the complete transient for the three signals.
- Save a copy of that. Keep in mind the tips in Activity 2.

- c. Using the magnifier tool, adjust the horizontal and vertical scale to show 5 cycles of the waveforms in steady state. You can also use the tools under *Axis* to set the X and Y ranges more accurately, or configure your print time and step time with the *Simulation Control* tool.
- d. Save a copy of that.

ACTIVIY 4 – TIME STEP

- a. Play with the *Time Step* under the *Simulation Control* tool. Get three sets of plots showing 5 cycles of the waveforms in steady state for:
 - i. A large step-size
 - ii. A reasonable step-size
 - iii. A very small step-size
- b. Save a copy of each set of curves.
- c. Compare the results and discuss on the processing burden of each approach.

ACTIVIY 5 – MEASUREMENTS

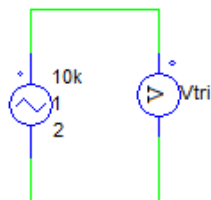
- a. Measure the mean value for each of the waveforms using *Analysis/Avg* adjusting the time span for the following cases:
 - i. 10 complete cycles of the steady-state waveforms.
 - ii. 2 cycles and a fraction of a third one of the steady-state waveforms.
 - iii. Several cycles (around 50) of the steady-state waveforms.

Tip: you can control precisely how many cycles will be considered using the tools under *Axis* to set the X and Y ranges accurately, or configure your print time and step time with the *Simulation Control* tool.

- b. Discuss on the differences among the three cases. What would be a good practice for measuring the mean value?
- c. Zoom in to get 5 complete cycles and measure the peak values for the inductor current waveform using the *Measure* tool and show the maximum values using *Mark Data Point* (*Measure* menu).
- d. Save a copy of the inductor current vs. time plot and measurements.

ACTIVIY 6 – FFT

- a. Build the following circuit in a new PSIM file and configure the triangular-wave voltage source as detailed on the table:



Parameter	Value
Frequency	10 kHz
Peak to peak voltage	1 V
DC offset	2 V

- b. Run the simulation and plot the voltage waveform.
- c. Perform the FFT of the waveform using *Analysis/Perform FFT*.
- d. Include the *Simulation Control* tool and repeat points b) and c) for:

- i. A larger *Total Time* (10 times or more)
 - ii. A smaller *Total Time* (10 times or more)
- e. Discuss about what you observe.
- f. Now, keep the *Total Time* fixed and change the step-size. Repeat points b) and c) for:
 - i. A step-size 10 times smaller
 - ii. A step-size 100 times smaller
- g. Discuss about what you observe.

EXPERIMENTAL WORK

ACTIVITY 1 – OSCILLOSCOPE AND PROBES CALIBRATION

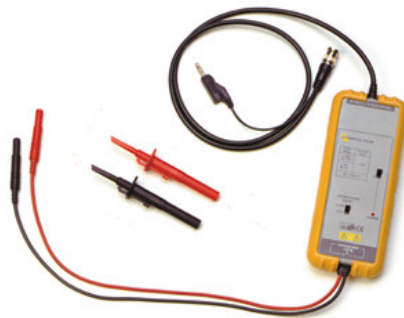
For the power electronics course we are going to use 3 kind of probes:



Passive probes

(x10 attenuation, 300V max)

Note: the ground clip is connected to the ground of the oscilloscope. Always make sure you connect all the ground clips of passive probes to the same point in the circuit.



Differential probes

(x20/x200 attenuation, 1400V max)

Uses external AC adaptor to power. Make sure to turn on (red LED should be on).

Voltage is measured from black lead to red lead (it can be inverted with no problem).



Current probes

(5A/30A range)

Uses external AC adaptor connected to the oscilloscope.

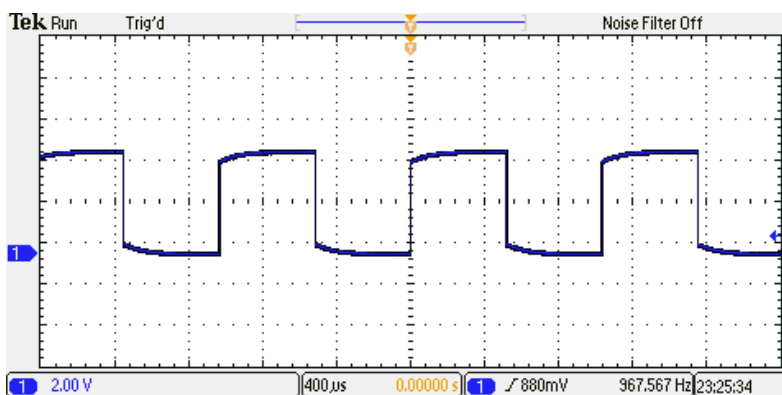
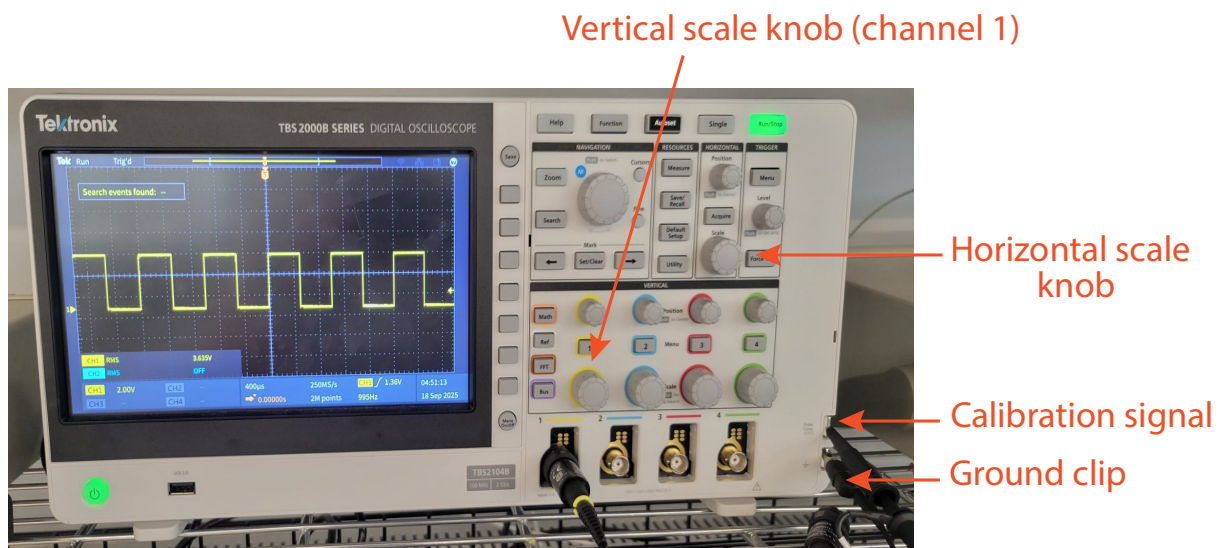
Has an arrow in the clamp that indicates the defined positive direction.

It can be inverted.

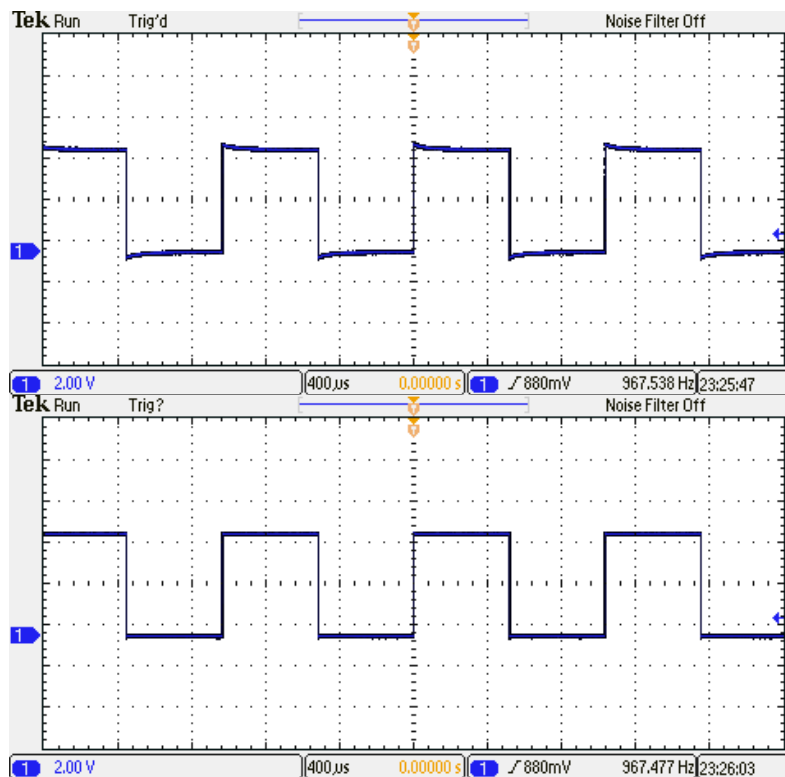
The following instructions in the guide below are to configure the probes. These steps must be followed every time you use the scope, otherwise the measurements may be wrong by orders of magnitude.

a. Passive probe calibration and setup:

- Connect the probe to the channel of the scope and connect the tip and ground to the calibration signal of the scope (a 1 kHz, 0-5 V, square wave).
- Using the channel menu, set the multiplier to the correct value (usually x10 for passive probes).
- Use the knob of the channel to set the vertical scale to the appropriate value to use the display completely while not clipping the signal.
- Using the horizontal scale knob, select the proper horizontal scale to display a couple of cycles of the calibration signal.
- The wave in the screen should look like a square wave, not rounded in the corners. If you see this effect, use a screwdriver to compensate the probe.



Undercompensated probe



Overcompensated probe

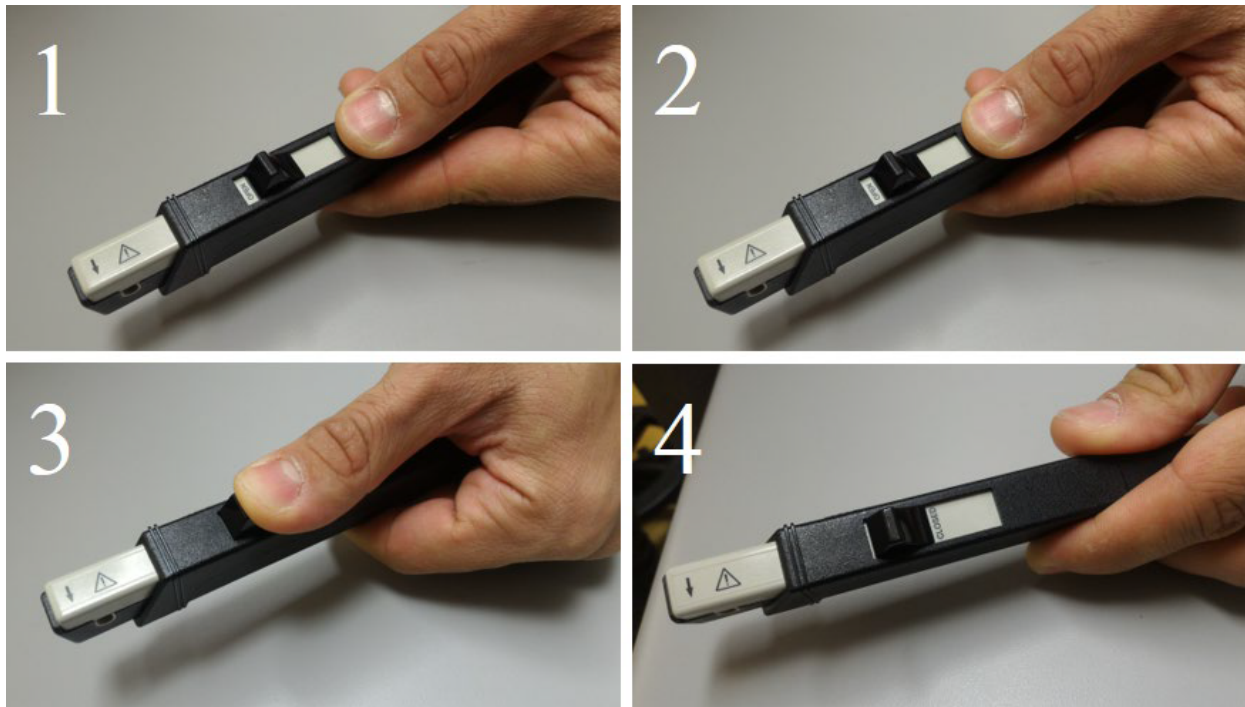
Correctly compensated probe

b. Differential probe calibration and setup:

- i. Connect the BNC side of the probe to the oscilloscope channel.
- ii. Connect the auxiliary power supply to the AC socket and turn the probe on.
- iii. Select the proper scale for the probe switch (usually x20 or x200)
- iv. Connect the tips of the probe to the calibration signal (a 1 kHz, 0-5 V square wave).
- v. Using the channel menu, set the multiplier to the correct value, matching what you configured in the probe
- vi. Use the knob of the channel to set the vertical scale to the appropriate value to use the display completely while not clipping the signal.
- vii. Using the horizontal scale knob, select the proper horizontal scale to display a couple of cycles of the calibration signal.

c. Current probe calibration and setup:

- i. Connect the probe to the oscilloscope and the auxiliary power supply to the scope.
- ii. Close the probe and lock it. Pay attention to the fact that the probe has two steps to close: closed and locked. For locking, you need to push the latch to the complete end, as shown in the pictures. If the probe is not locked, the probe will display a red light to warn the user and a message will be displayed in the scope. Pay attention to this since it will cause measurement errors!



- iii. Select the scale for the probe by pressing the “range” button.
- iv. Using the “Degauss/AutoZero” button, fix the offset of the probe.
- v. The probe is ready to use when no red light is displayed in the connector. Do not take measurements if there is a red light in the probe (as shown in the picture) since it will produce wrong readings. If the probe needs to be degaussed, it will need to be removed from the circuit, locked and then calibrated.
- vi. The current probe is able to change the multiplier and units automatically.
- vii. The probe clamp has an arrow indicating the direction of the positive current flow, as displayed in a circuit schematic. Connect the probe pointing in the direction defined as positive, or the measurement will be inverted.



GUIDELINES FOR PRINTING WAVEFORMS FROM OSCILLOSCOPE

- a. **Selection of relevant information.** When taking an oscilloscope capture, be sure that the screen has all the information that is relevant for what you want to show, and hide any other piece of information that is not relevant.
 - i. Hide any channel that is not needed.

- ii. Include the measurements of any parameter that is relevant for the activity and hide what is not important.
 - iii. Hide any panel or zoom window that is not required
- b. **Trigger: single.** The digital oscilloscope that we use shows an overlap of a number of samples of the same signal at every time. That is observed as a wider waveform with different levels of brightness. However, when we take a capture we usually want the waveform to look *cleaner* than that. If we use *single* shot instead of *stop*, the waveform that we get is a single sample of our signal.
- c. **Proper distribution of waveforms.** Separate the waveforms that you do not want to overlap and adjust the vertical scales accordingly. Use the whole oscilloscope screen.
- d. **Saving to USB flash-drive (not pictures).** Print the captures to the flash-drive plugged into the oscilloscope USB port (<2 GB preferred to avoid compatibility issues). Photographs of the oscilloscope's screen are not acceptable on the reports

ACTIVITY 2 – TIME DOMAIN CAPTURES AND MEASUREMENTS

- a. Connect a regular probe of the oscilloscope to the output of the function generator. Configure the function generator to output a triangular wave with the following characteristics:

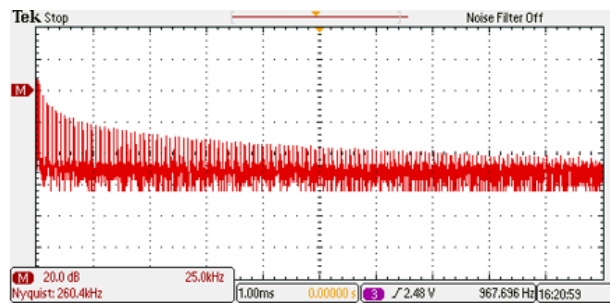
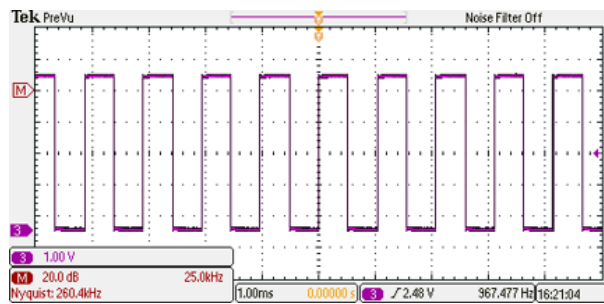
Parameter	Value
Frequency	10 kHz
Amplitude	500 mV
DC offset	2 V

- b. Configure the scope to properly display the waveform. Use the trigger to synchronize the waveform (the level should be adjusted so the signal is steady). Using the measurements menu, measure: frequency, peak-to-peak amplitude and mean value. Are the values measured by the scope accurate?
- c. Now, reduce the amplitude of the signal to 10 mV, keeping the 2 V DC offset. Repeat the measurements. Are the measurements accurate now?
- d. Now, configure the oscilloscope to remove the DC component (found in the channel menu of the scope). Repeat the measurement. Is the ripple is easier to observe now? Are the measurement more accurate?
- e. Discuss the previous points.

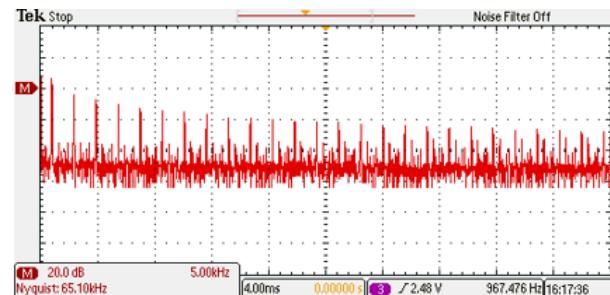
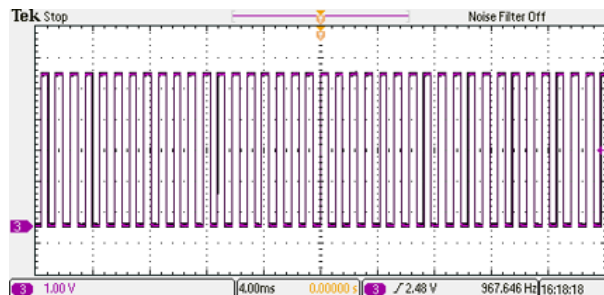
ACTIVITY 3 - DOMAIN CAPTURE AND MEASUREMENT

- a. Using the same configuration as in Activity 2 a., use the math function to visualize the Fast Fourier Transform of the signal in the screen. What frequency components do you expect to find in the FFT?
- b. Adjust the time scale using the horizontal scale knob in order to achieve good frequency resolution while avoiding aliasing effect as shown in the examples below:

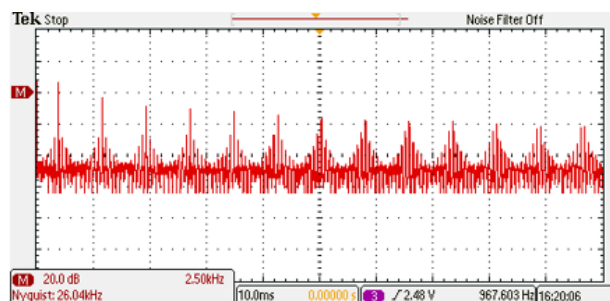
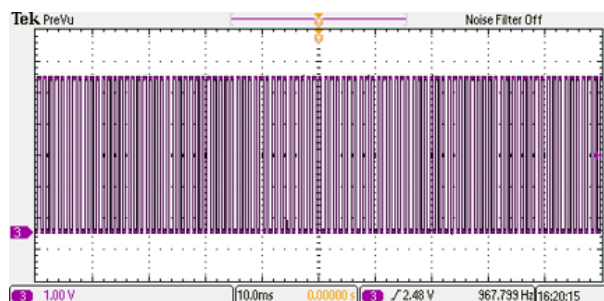
- i. Time-scale too small, low frequency resolution:



- ii. Larger time-scale, higher frequency resolution:



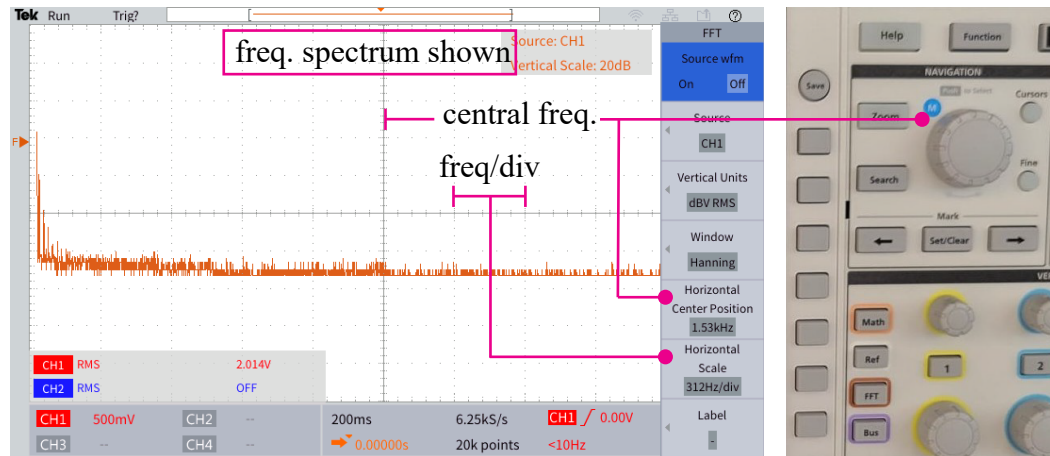
- iii. Even larger time-scale, higher frequency resolution, but creating aliasing effect:



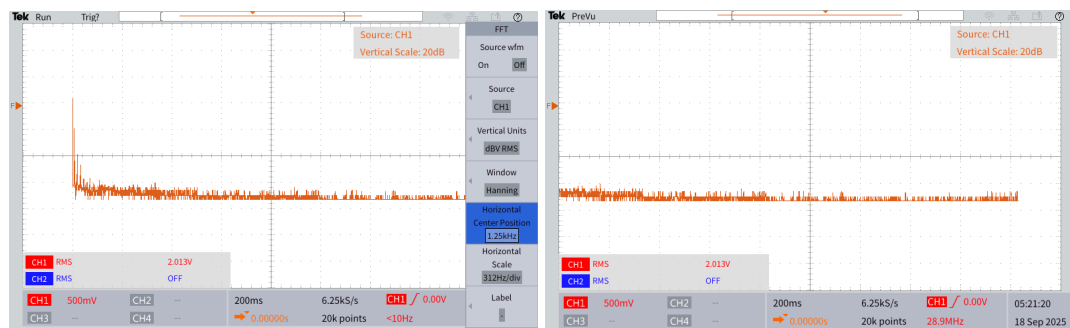
To do so, use the FFT menu and the multipurpose knobs to adjust the horizontal scaling of the FFT so that the appropriate components are displayed. Follow the guidelines shown below:

- i. There is a knob “M” that controls the displacement of the frequency spectrum, by displacing the central frequency, and controls the frequency per division parameter.

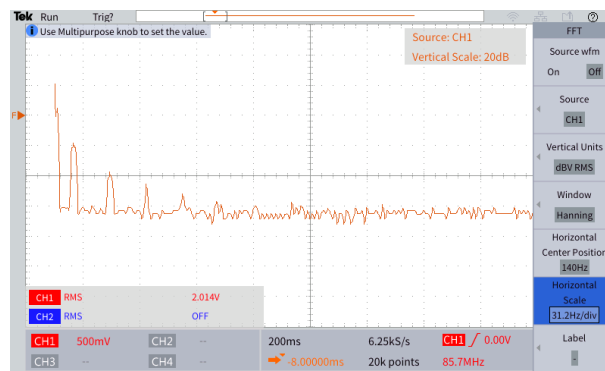
full frequency spectrum



ii. By using knob “M”, it is possible to change the central frequency being shown.



iii. By using “knob “M”, it is possible to change the “freq/div” of the plot.



- Capture the results in time and frequency domains in two separate images. Make sure not to change any configuration parameters in between captures.
- Change the frequency of the signal to 20 kHz and observe the difference in the harmonics. Capture the results.
- Change the signal to a sine wave and observe the differences. Capture the results.
- Discuss the previous point.